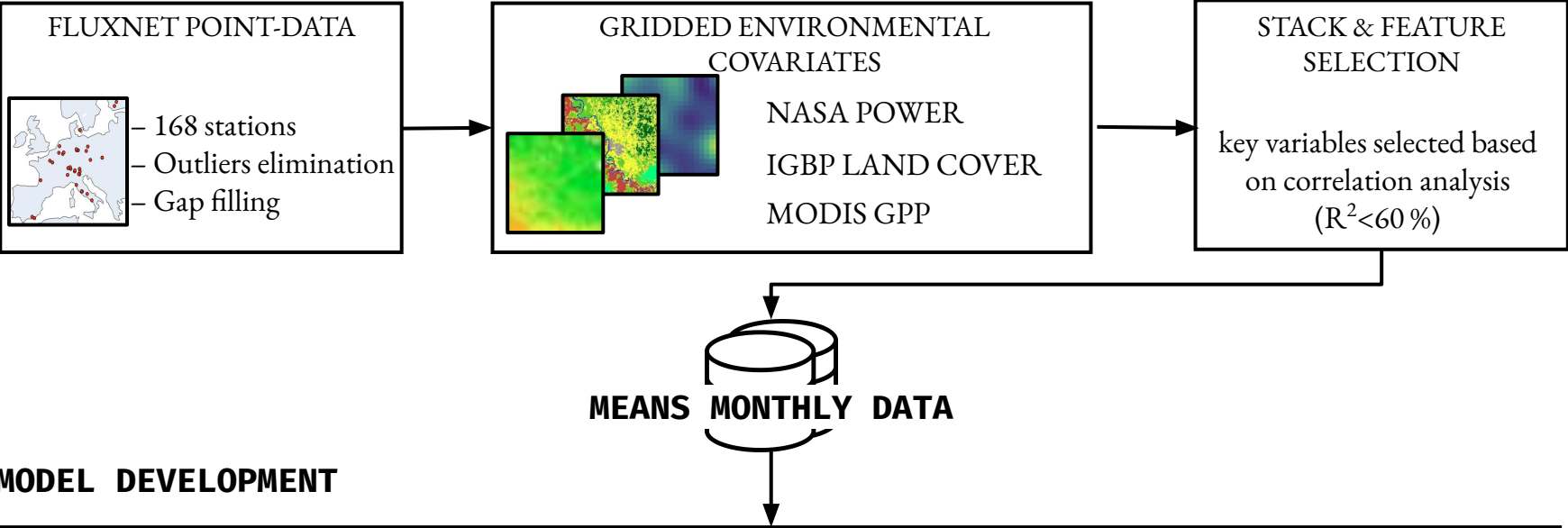


DATA PREPARATION

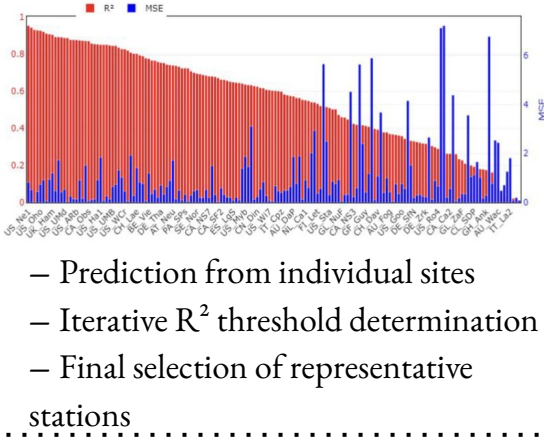


MODEL DEVELOPMENT

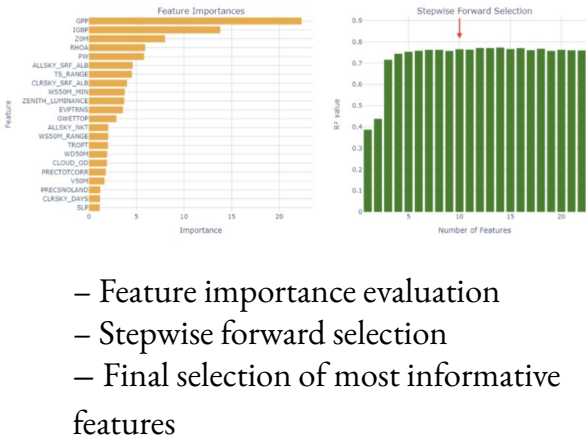
PRE-TRAINING
STAGE: OPTIMAL
SET OF THE DATA

- CatBoost for regression
- Optuna framework for hyperparameter optimization

REPRESENTATIVE STATIONS



MOST INFORMATIVE FEATURES



FINAL DATASET

TRAINING STAGE

Station US_Oho ($R^2 = 0.92$, MSE = 0.91), DBF

TIME-SHIFT ROBUSTNESS

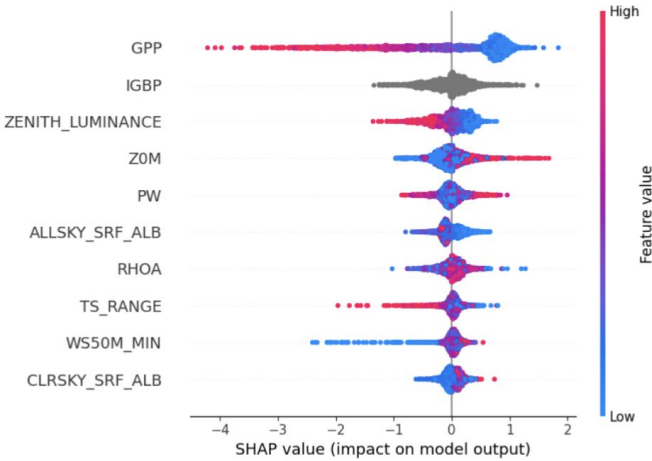
- 80/20 % train/test chronological split
- R^2 (0.76) and MSE (1.05) average accuracy assessment
- Biome-wise accuracy assessment
- Stations-wise accuracy assessment

SPATIAL CONSISTENCY

- 80/20 % train/test chronological split except station RU-Fyo, located in Tver Oblast, Russia
- R^2 (0.58) and MSE (0.45) validation accuracy assessment

APPLICATIONS

EXPLAINING COVARIATES IMPACT
WITHIN MODELLING SETTING



INFERENCE OVER TEST-REGION

